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Method of measurement of interfacial tension (IFT) of two immiscible fluids at reservoir conditions: an in situ gas buffered injection

Abstract

A method for determining interfacial tension of a hydrocarbon in a brine fluid, the method including injecting a first brine fluid into a test cell, the first brine fluid having an initial ionic composition, injecting a hydrocarbon fluid into the test cell, contacting the hydrocarbon fluid with the first brine fluid, forming a droplet, measuring the interfacial tension of the hydrocarbon fluid in contact with the first brine fluid, at least partially displacing the first brine fluid with an inert gas, measuring a ionic composition salinity of the displaced first brine fluid in an ionic chromatograph, and comparing the measured ionic composition salinity to the initial ionic composition.

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Background/Summary

BACKGROUND

(1) Water flooding with brine or seawater composition is often used for secondary recovery processes in hydrocarbon-bearing reservoirs. Enhancing oil recovery in naturally fractured reservoirs by injecting a chemistry-optimized water has been widely investigated and gained increasing interest in the last decade. This interest is mainly driven by the attractive economics of such technology because of the existing water flooding facilities. Numerous multi-scale research efforts were initiated in order to capture the impact of modified injected water composition on reservoir rocks wettability and the increase of reservoirs oil recovery.

(2) Fluid/fluid interface behavior is an important parameter in the understanding of the effect of the injected brine ionic composition. The interfacial tension (IFT) between two immiscible fluids is a physical measurement, much needed in the oil industry in general and enhanced oil recovery, in particular. The measurement of interfacial tension between hydrocarbon and brine fluid is mainly used to assess the effect of injected brine composition, additives (surfactant, polymers, alkali . . .) and also the activity of the interface at defined temperature and pressure.

(3) Interfacial tension measurement of hydrocarbon/brine fluid, for example, is based on an optical method. A hydrocarbon drop with vertical symmetry is generated from a vertical needle, surrounded by a brine fluid. The hydrocarbon drop shape results from a balance between gravity

forces, buoyancy, pressure caused by the curvature interface and interfacial tension forces.

Interfacial tension is then calculated from the hydrocarbon drop using the Laplace equation.

(4) The above process requires extensive cleaning to avoid contamination, which is also time consuming when it comes to testing multiple brines with different compositions, additives, and temperature and pressure conditions.

(5) The prior art describes interfacial tension measurements by using a static method that requires multiplication of single runs. Screening the effect of additives and fluids compositions on the interfacial properties between two immiscible fluids is usually performed by completing a measurement for one set of fluids followed by a drainage, a thorough cleaning, injection and then IFT measurement for the following set of fluids.

(6) The current practices are based on measuring IFT of fluids with hydrocarbon on separate single runs. This method requires intensive cleaning, time and is a source of experimental fluctuations. This will add uncertainties associated with each individual run and will not emulate the real interfacial changes occurring at the interfaces of brines with additives and/or modified ionic composition.

SUMMARY

(7) In one aspect, embodiments disclosed herein relate to a method for determining interfacial tension of a hydrocarbon in a brine fluid. The method may include injecting a first brine fluid into a test cell, the first brine fluid having an initial ionic composition, injecting a hydrocarbon fluid into the test cell, contacting the hydrocarbon fluid with the first brine fluid, forming a droplet, measuring the interfacial tension of the hydrocarbon fluid in contact with the first brine fluid, at least partially displacing the first brine fluid with an inert gas, measuring a ionic composition salinity of the displaced first brine fluid in an ionic chromatograph, and comparing the measured ionic composition salinity to the initial ionic composition.

(8) In another aspect, embodiments disclosed herein relate to a method for determining interfacial tension of a hydrocarbon in a brine fluid. The method may include injecting a first brine fluid into a test cell, the first brine fluid having an initial ionic composition, injecting a hydrocarbon fluid into the test cell, contacting the hydrocarbon fluid with the first brine fluid, forming a droplet, measuring the interfacial tension of the hydrocarbon fluid in contact with the first brine fluid, at least partially displacing the first brine fluid with an inert gas, measuring a ionic composition of the displaced first brine fluid in an ionic chromatograph, and comparing the measured ionic composition to a second initial ionic composition of a second brine fluid.

(9) In another aspect, embodiments disclosed herein relate to a system useful for determining interfacial tension of a hydrocarbon in a brine fluid. The system may include a test cell configured to enclose a hydrocarbon fluid disposed in a first brine fluid, a first brine fluid tank configured to hold a volume of the first brine fluid, the first brine fluid having an initial ionic composition, a second brine fluid tank configured to hold a volume of a second brine fluid, the second brine fluid having a second initial ionic composition, at least one pump fluidly coupled to the test cell and the first and second brine fluid tanks, and a control system communicably coupled to the pump.

(10) In another aspect, embodiments disclosed herein relate to a process for determining interfacial tension of a hydrocarbon in a brine fluid. The process may be performed in a system including a first pump (126) fluidly connected via a valve V0 to a first brine fluid tank (120a) holding a volume of a first brine fluid and via a valve V2 to a second brine fluid tank (120b) holding a volume of a second brine fluid, a second pump (125) fluidly connected via a valve V11 and a valve V12 to an inert gas tank (116) holding a volume of inert gas and via the valve V11 and a valve V13 a hydrocarbon tank (115) holding a volume of hydrocarbon fluid, and a test cell (105) configured to measure the interfacial tension of the hydrocarbon fluid. The test cell may be fluidly connected via a valve V4 and valves V7, V8, and V10 to the first brine fluid tank, via a valve V5 and the valves V7, V8, and V10 to the second brine fluid tank, via a valve V14 to the inert gas tank, and via a valve V15 and a valve V19 to the hydrocarbon fluid tank. The process may include opening valves

V0, V1, V4, V7, V8, and V10, injecting the first brine from the first brine tank using the first pump, filling the test cell with the first brine fluid, heating the test cell with a heating coil, stopping injecting the first brine fluid when a pressure and a temperature of the test cell have reached reservoir conditions, and closing valves V0, V1, V4, V7, V8, and V10, opening valves V11, V13, V15, and V19, injecting the hydrocarbon fluid from the hydrocarbon fluid tank into the test cell using the second pump, closing the valves V11, V13, V15, and V19, stopping injecting the hydrocarbon fluid, and measuring the interfacial tension of the hydrocarbon droplet in the brine fluid.

(11) Other aspects and advantages will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic illustration of an example implementation of the system according to one or more embodiments disclosed herein.

(2) FIG. 2 is a flowchart that illustrates an example method according to one or more embodiments disclosed herein.

(3) FIG. 3 is a flowchart that illustrates an example method according to one or more embodiments disclosed herein.

(4) FIG. 4 is an illustration of IFT measured using the system and method according to one or more embodiments disclosed herein.

(5) FIG. 5 is a schematic illustration of an example control system according to one or more embodiments disclosed herein.

DETAILED DESCRIPTION

(6) Embodiments disclosed herein relate to methods and systems for improving IFT measurement by using an inert gas, such as nitrogen, to displace and replace brine fluids inside a high pressure high temperature (HPHT) chamber at experimental conditions. The sequential replacement is completed when the ionic composition of the drained fluid matches the initial injected fluid. This may ensure a consistent screening for wettability alteration modifiers and an optimized experimental protocol. As used herein, “match” can refer to equivalent compositions and compositions that are substantially the same, such as a drained fluid having an ionic composition that is more than 90% that of the initial injected fluid, an ionic composition that is more than 95% that of the initial injected fluid, an ionic composition that is more than 98% that of the initial injected fluid, or an ionic composition that is more than 99% that of the initial injected fluid.

(7) The present disclosure describes implementations of systems and method for determining interfacial tension (IFT). In some aspects, such implementations include an in-situ dynamic measurement method to determine an ionic composition gradient effect on a droplet of a hydrocarbon fluid. Example implementations include a test cell that is used for measurements at various conditions (for example, at standard conditions, high temperature/high pressure conditions, or both). In some aspects, in-situ brine liquids may be circulated to the test cell to assess an impact of an ionic composition gradient on oil recovery. In some aspects, the circulation of multiple brines (for example, serially) may be controlled by measuring a parameter of the brines (for example, by using ionic chromatography) and using the measurement to control brine replacements inside the test cell. In some aspects, such implementations may enable a realistic measurement of IFT caused by an ionic composition gradient and various determining parameters (for example, crude oil composition, reservoir pressure and temperature, and other parameters). One such example of an ionic composition of the hydrocarbon fluid disclosed herein is the salinity of the hydrocarbon fluid. The salinity of the hydrocarbon fluid may change depending on the brine fluid used for testing.

(8) Accordingly, a new experimental method to measure in-situ interfacial tension between hydrocarbon fluids and brine fluids at reservoir conditions, of different compositions injected successively is disclosed herein. The method may be buffered by an inert gas, such as nitrogen. Interfacial tension (IFT) of a first brine fluid may be measured and then nitrogen gas may be injected gradually inside the cell to purge the first brine fluid. The drained brine fluid may then be diverted to an in-line ionic chromatography analyzer (IC) to determine the ionic composition. Nitrogen gas injection may stop as soon as the first brine fluid is completely drained. A second brine fluid may then be gradually introduced to the cell while maintaining temperature and pressure conditions. A third, fourth, fifth, etc. brine fluid may also be introduced in this fashion (inert gas displacement followed by brine fluid injection). This approach may provide a contamination-free, time efficient process in measuring interfacial tension between hydrocarbon fluid and injected brine fluid.

(9) The interfacial tension between two non-miscible liquids, or between a liquid and an ambient fluid, is a physical magnitude that is much used in various scientific and technical fields. It represents the energy brought into play by the forces of intermolecular cohesion on an interface between a liquid and an ambient medium. An ambient medium is a medium, which is also fluid, into which the liquid considered is located. The liquid and the ambient medium are non-miscible. Interfacial tension is expressed in terms of force per unit of length, for example in Newton/m or millinewton/m.

(10) Calculating the value of the interfacial tension between a liquid and an ambient medium is generally known, and operates on the basis of optical measurements of the shape and dimensions of a drop of liquid hydrocarbon in the ambient medium and clinging to the end of a hollow needle used to form the drop. The drop takes a shape that results from the equilibrium among the forces of gravity, buoyancy, the forces of interfacial tension and the pressure due to the curvature of the interface. It can thus be seen that, for two given liquids, the separation of the drop from its own medium depends on the interfacial tension of the radius of the hollow needle and on the angle of contact. This angle is directly related to the material constituting the needle.

(11) By analyzing the shape and dimensions of the drop, it is possible to determine the value of the interfacial tension in an absolute way. There may be a relationship between the value of the interfacial tension and the geometrical parameters characterizing the shape of the drop.

(12) Turning now to the figures, FIG. 1 is a schematic illustration of one embodiment of the IFT measurement system **100**. As shown, the system **100** includes a test cell **105** that encloses a liquid hydrocarbon dropper **110**. In some aspects, the liquid hydrocarbon drop deposited by the liquid hydrocarbon dropper **110** is a crude oil sample (or other hydrocarbon liquid sample).

(13) The test cell **105**, in some aspects, may be a contact angle goniometer that is operable (for example, through imaging devices and a microprocessor based software system) to measure interfacial tension between the hydrocarbon fluid sample and the liquid brine. In some aspects, the test cell **105** may be a DSA100, KRUSS GmbH or IFT-10, Corelab.

(14) Using these devices, different interfacial tension between hydrocarbon fluid drops and liquid brine may be measured.

(15) The IFT measurement, in some aspects, may help determine an effectiveness of secondary or tertiary production processes, such as water flooding. For example, water flooding, generally, refers to the process of injecting water (or water-based liquid such as brine) into a reservoir to increase reservoir pressure and thereby increase hydrocarbon production. Thus, the wettability of a reservoir formation and mobility of the hydrocarbon fluid—whether such formation is water-wet or oil-wet—and how water flooding can effect such wettability and hydrocarbon mobility, as determined through interfacial tension, may affect a choice or operation of stimulation.

(16) The system **100** includes a hydrocarbon fluid sample tank **115** that is fluidly coupled to the test cell **105** though a valve **V15** (for example, modulating or shut-off). Although shown as a single tank **115**, there may be multiple different tanks **115** fluidly coupled to the test cell **105** through a

single or multiple valves **V15**. Each tank **115** may enclose a similar or different type of hydrocarbon fluid sample (for example, crude oil from different formations).

(17) As illustrated, the system **100** also includes a first brine fluid tank **120a** that is fluidly coupled to the test cell **105** through valve **V4**. A second brine fluid tank **120b** is fluidly coupled to the test cell **105** through valve **V5**. A third brine fluid tank **120c** is fluid coupled to the test cell **105** through valve **V6**. Although shown as three tanks, there may be greater or fewer than three tanks fluidly coupled to the test cell **105** through a single or multiple valves in a similar manner as illustrated. Each tank may enclose a similar or different type of brine (for example, each having a different salinity, different conductivity, or other property).

(18) One or more pumps **125** is also fluidly connected to the hydrocarbon fluid sample tank **115**. A second pump **126** is also fluidly connected to the brine tanks **120a-c**. Both pumps may be operable to circulate the hydrocarbon fluid and brine fluids from the respective tanks **115** and **120a-c**. As illustrated, the pump **125** is fluidly connected to the tank (or tanks) **115** through valves **V11** and **V13**, while the pump **126** is also fluidly connected to the tanks **120a-c** through valves **V0**, **V1**, **V2**, and **V3**.

(19) The one or more pumps **125** may also be fluidly connected to an inert gas tank **116** through valves **V11** and **V12**. The inert gas tank **116** may be fluidly connected to the test cell **105** through valve **V14**. The inert gas held in inert gas tank **116** may be useful for flushing the test cell **105** after an IFT measurement is performed. The inert gas may be injected into the test cell at reservoir pressure, and heating element **106** may heat the inert gas to reservoir conditions in the test cell **105**. Accordingly, the test cell may be kept at reservoir conditions during filling, testing, and purging. Additionally, the flow lines running from brine tanks **120a-120c** may be a heated line equipped with an electrical trace, or other similar device, to deliver the brine to the test cell **105** at, or near, reservoir temperature.

(20) Pumps **125** and **126**, as well as the tanks **115**, **116**, and **120a-c** may be connected such that the inert gas inlet may be located at the top of the test cell **105**, while the liquid inlets for the one or more brines and hydrocarbon fluid may be located at the bottom of test cell **105**. Further, the pumps **125** and **126**, and the gas flowrate, may be controlled to minimally disturb the liquids. For examples, the pumps may be operated such that the fluids and gases are slowly pushed rather than being rapidly pushed causing a blowout, splashing, mixing, etc.

(21) The volume of inert gas may depend on the test experimental conditions such as temperature, pressure, and brine composition. Injecting of the inert gas may be controlled by pump **125** and the back pressure regulator (BPR) **130**. The inert gas injection may continue until the brine is completely drained through valve **V9** (i.e., no additional fluids are collected from valve **V9**), or until after ionic chromatography test is performed with the excess inert gas being vented by valve **V17**. BPR **130** may be equipped with one or more valves (**V16**, **V17**, and **V18**) to control the pressure of the inert gas as it is injected into test cell **105**, or allow the removal of inert gas when brine is pumped into test cell **105**.

(22) As illustrated in FIG. 1, the system **100** also includes a drain system for removing a brine/hydrocarbon sample after IFT testing. Valve **V10** (for example, modulating or shut-off) may be fluidly coupled to an outlet of the test cell **105** and a filter **135** through valve **V9**. Once filtered, the brine may be fed through valves **V21** and **V20** to an ionic chromatography (IC) unit **140**. The IC **140** measures the level of dissolved salts, or salinity, of the brine that is removed from the test cell **105**. By comparing the measured dissolved salts of the drained brine to the known level of dissolved salts in the fresh brine, it may be possible to determine if the test cell **105** has been completely cleaned of residual hydrocarbon.

(23) System **100** also includes a control system (for example, microprocessor based, electromechanical, pneumatic, or other form of control system (not illustrated)). The control system may send commands to, and receive information from, the pumps. For example, the control system may send commands to the pump(s) to start or stop, or slow down or speed up, or both. The

pump(s) may send feedback to the control system, such as speed, flow rate, frequency, or a combination. In some aspects, the control system may include or be controllably coupled to a variable frequency drive connected to the pump motor.

(24) The control system may also be communicably coupled to send commands to, and receive information from, the valves **V0-V21**. For example, the control system may send commands to the valves **V0-V21** to open fully or close fully, modulate toward open or modulate toward close. The valves **V0-V21** may send feedback to the control system, such as status (open or close), percent open, or a combination.

(25) The control system may also be communicably coupled to send commands to, and receive information from, the test cell **105**. For example, the control system may send commands to the test cell **105**, such as, to perform IFT measurement or heat the test cell with heating element **106**.

(26) The control system may also be communicably coupled to receive information from the IC **140**. For example, the control system may send a signal to the IC **140** to perform the chromatography, and may receive feedback from the IC **140** in the form of a measured ionic composition.

(27) Using the system **100** as illustrated in FIG. **1**, the following process steps for successive IFT measurements may be performed.

(28) The control system may turn on pump **126**, open valves **V0**, **V1**, **V4**, **V7**, **V8**, and **V10**. The pump **126** may inject a first brine from brine tank **120a**, filling test cell **105** with the first brine fluid. The heated line from the brine tank **120a** to the test cell **105**, and/or the heating element **106** may heat the brine and test cell to reservoir conditions, and the pump **126** may provide the pressure to keep the first brine at reservoir pressure. When pressure and temperature have reached reservoir conditions and stabilized, the control system may close valves **V10**, **V8**, **V7**, **V4**, **V1**, **V0**, and stop pump **126**. These operations may happen simultaneously, or sequentially in any order.

(29) After the control system closes valves **V10**, **V8**, **V7**, **V4**, **V1**, **V0**, and stops pump **126**, the control system may turn on pump **125**, open valves **V11**, **V13**, **V15**, and **V19** to generate an inverted pendant drop of the hydrocarbon fluid from tank **115** in test cell **105**. The control system may then close valves **V19**, **V15**, **V13**, and **V11**, stop pump **125**, and start measuring interfacial tension (IFT) versus time of the droplet within the test cell **105**.

(30) After the IFT test is performed, the control system may start pump **125**, open valves **V11**, **V12**, **V14**, and **V16** to inject inert gas from inert gas tank **116** into the test cell **105**. The control system may also open valves **V10** and **V9** to drain the first brine fluid from the test cell **105** to the filter **135**. During inert gas injection, the control system may use pressure adjustment valve **V18** to maintain a stable reservoir pressure within the test cell **105** during the purge process. Pressure adjustment valve **V18** may be at least partially opened to let a portion of the inert gas leave the system, thereby stabilizing the pressure generated by pump **125**. Valve **V18** may be controlled by the BPR **130** and the control system to maintain the test cell pressure constant during the displacement process. If the vessel pressure is higher than the test pressure, valve **V18** will release pressure to maintain the equilibrium. If the vessel pressure is lower than test pressure, **V18** will close, causing pump **125** to pressurize the test cell to maintain the equilibrium pressure.

(31) Once the first brine fluid is fully drained from test cell **105**, the control system may stop pump **125** and close valves **V16**, **V14**, **V12**, **V11**, **V10**, **V9** and **V20**. The drained first brine fluid may be collected and filtered in filter **135**. After filtering, the control system may open valve **V20** to allow the filtered brine fluid to enter the IC **140** for analysis. The IC **140** will measure the ionic composition of the drained brine fluid and compare the measured ionic composition of the drained brine fluid against a known ionic composition of fresh brine fluid. If the measured ionic composition matches the known ionic composition of the fresh brine fluid, the control system will indicate that the test cell **105** has been completely cleaned. If the measured ionic composition does not match the known ionic composition of the fresh brine fluid, the control system will indicate that the test cell **105** has not been completely cleaned and repeat the above steps to inject the first brine

fluid into the test, and then purge the first brine fluid with inert gas, without injecting the hydrocarbon fluid. This process is repeated until measured ionic composition matches the known ionic composition of the fresh brine fluid.

(32) Once the control system has determined that the test cell **105** has been completely cleaned, the control system will begin testing the second brine fluid by starting pump **126** and opening valves **V0**, **V2**, **V5**, **V7**, **V8**, and **V10** to fill the test cell **105** with the second brine fluid. The steps of injecting the hydrocarbon drop, measuring the IFT, and draining the test cell may then be repeated. Additionally, the above process may be repeated for the third brine fluid, as illustrated, and may start again with the first brine fluid, or proceed with a fourth, fifth, sixth, etc. brine fluid (not illustrated).

(33) During brine fluid injection into test cell **105**, the control system may open valve **V16** and vent valve **V17** to allow the release of inert gas from the test cell **105**. This release of inert gas may maintain the test cell **105** at isobaric conditions. During injection of inert gas and draining of the brine fluid from test cell **105**, vent valve **V17** may be closed.

(34) FIG. **2** is a flowchart that illustrates an example method **300** for measuring IFT between the hydrocarbon fluid sample and the brine fluid with the system **100**. In some aspects, by determining one or more IFT between the hydrocarbon fluid sample and two or more different brine fluids, a surface tension change may be more accurately and easily determined in a single test set up with the test cell **105**. Method **300** may begin at step **302**, which includes injecting a brine fluid into the test cell to fill the cell.

(35) Method **300** may continue at step **304**, which includes injecting a hydrocarbon fluid to the test cell. Step **306** includes measuring IFT between the hydrocarbon fluid and the brine fluid. Step **308** includes displacing the brine fluid in the test cell with an inert gas. For example, once the IFT has been measured in step **306**, the brine fluid may be drained by injecting inert gas so that the test cell may be cleaned and avoid contaminating the next brine fluid.

(36) Method **300** may continue with step **310**, which includes collecting the drained brine fluid. The collected fluid may also be filtered in a filter. Step **312** may include measuring the ionic composition of the drained brine fluid. For example, the ionic chromatography unit may measure ionic composition. Such measured values may be sent to the control system from the IC. The control system may compare the measured values to a known (or measured) value of the same property of the fresh brine fluid.

(37) Method **300** may continue at step **314**, which includes a determination of whether the measured property (for example, salinity) of the displaced brine fluid matches the known or measured same property of the fresh brine fluid. For instance, when the measured value of the property of the displaced brine fluid matches (or substantially matches, within 1%) the known value of the property of the fresh brine fluid, the method may proceed at step **316** and the control system may stop the flow of inert gas to the test cell **105**. If the determination in step **314** does not match the property of the fresh brine fluid, then method **300** continues at step **315**, where additional brine fluid is circulated to the test cell, and steps **308**, **310**, **312**, and **314** are repeated until the measured property matches the known fresh property. After the inert gas is stopped in step **316**, the method may continue at step **318** and determine if another brine fluid, or another hydrocarbon, is to be tested. If another brine fluid is to be tested, the method may start over at step **302**, otherwise the method ends.

(38) FIG. **3** is a flowchart that illustrates an example method **350** for measuring IFT between the hydrocarbon fluid sample and the brine fluid with the system **100**. In some aspects, by determining one or more IFT values between the hydrocarbon fluid sample and two or more different brine fluids, a surface tension and oil mobility may be more accurately and easily determined in a single test set up with the test cell **105**. Method **350** may begin at step **352**, which includes injecting a brine fluid into the test cell to fill the cell.

(39) Method **350** may continue at step **354**, which includes injecting a hydrocarbon fluid to the test

cell. Step **356** includes measuring IFT between the hydrocarbon fluid and the brine fluid. Step **358** includes displacing the brine fluid in the test cell with an inert gas. For example, once the IFT has been measured in step **356**, the brine fluid may be drained by injecting inert gas so that the test cell may be cleaned and avoid contaminating the next brine fluid.

(40) Method **350** may continue with step **360**, which includes collecting the drained brine fluid. The collected fluid may also be filtered in a filter. Step **362** may include measuring the ionic composition of the drained brine fluid. For example, the ionic chromatography unit may measure ionic composition. Such measured values may be sent to the control system from the IC. The control system may compare the measured values to a known (or measured) value of the same property of the fresh brine fluid.

(41) Method **350** may continue at step **364**, which includes a determination of whether the measured property (for example, salinity) of the displaced brine fluid matches the known or measured same property of a fresh second brine fluid. For instance, when the measured value of the property of the displaced brine fluid matches (or substantially matches, within 1%) the known value of the property of the fresh second brine fluid, the method may proceed at step **366** and the control system may stop the flow of inert gas to the test cell **105**. If the determination in step **364** does not match the property of the fresh second brine fluid, then method **350** continues at step **365**, where the second brine fluid is circulated to the test cell, and steps **358**, **360**, **362**, and **364** are repeated until the measured property matches the known fresh property. After the inert gas is stopped in step **366**, the method may continue at step **368** and determine if another brine fluid, or another hydrocarbon, is to be tested. If another brine fluid is to be tested, the method may start over at step **352**, otherwise the method ends.

(42) Tables 1, 2 and 3 are comparison of ionic composition of the initial and the drained fluids, respectively (seawater, diluted seawater and deionized water).

(43) TABLE-US-00001 TABLE 1 Ionic composition of initial and drained seawater
Initial Seawater Drained Seawater Ions (mg/L) (mg/L) Calcium 690 702 Chloride 42077 41516
Magnesium 2027 2032 Sodium 17919 18323 Sulfate 3983 4151 TDS 66696 66724

(44) TABLE-US-00002 TABLE 2 Ionic composition of initial and drained diluted seawater
Initial Diluted Drained Diluted Ions Seawater (mg/L) Seawater (mg/L) Calcium 327 339 Chloride 15544 16350
Magnesium 954 975 Sodium 9371 10935 Sulfate 1965 2084 TDS 28161 30683

(45) TABLE-US-00003 TABLE 3 Ionic composition of initial and drained deionized water
Initial Deionized Drained Deionized Ions Water (mg/L) Water (mg/L) Calcium <10 <10 Chloride <5 <5
Magnesium <10 <10 Sodium <10 <10 Sulfate <10 <10 TDS — —

(46) Once the properties match, the method **350** may continue at step **368**, which includes a determination of whether another brine fluid (for example, a second brine fluid, a third brine fluid, a fourth brine fluid, and so on) is to be tested. For example, if multiple (for example, more than two) brine fluids are to be tested, then method **350** may continue back at step **352**. Otherwise, method **350** may end.

Example

(47) IFT measurements of three (3) crude oil/brine fluid systems were performed at 70° C. and 2000 psi. Successive replacement of brine fluid was performed using nitrogen gas to displace each brine fluid at isobaric and isothermal conditions. The following fluids were used: Brine 1: Seawater; Brine 2: Diluted seawater; Brine 3: Deionized water; Hydrocarbon: Dead crude oil.

(48) Drained brine was collected, filtered, and the ionic composition analyzed and compared with the initial injected brine compositions. This process was repeated with brine 1, 2, and 3.

(49) FIG. 4 illustrated the interfacial tension measurements between crude oil and the three different brines (seawater, diluted seawater and deionized water) replaced successively using the proposed method of nitrogen gas buffering. As seen, while IFT for deionized water starts high, above 13 mN/m, both the diluted seawater and seawater samples reached an equilibrium value higher than deionized water at t=500 s.

- (50) FIG. 5 is a schematic illustration of an example controller **400** (or control system) for an IFT measurement system. For example, the controller **400** can be used for the operations described previously, for example as or as part of a control system that performs one or more steps of method **300** described in FIG. 3. For example, the controller **400** may be communicably coupled with, or as a part of, the control system discussed in related to FIG. 1.
- (51) The controller **400** may include various forms of digital computers, such as printed circuit boards (PCB), processors, or digital circuitry. Additionally, the system can include portable storage media, such as, Universal Serial Bus (USB) flash drives. For example, the USB flash drives may store operating systems and other applications. The USB flash drives can include input/output components, such as a wireless transmitter or USB connector that may be inserted into a USB port of another computing device.
- (52) The controller **400** may include a processor **410**, a memory **420**, a storage device **430**, and an input/output device **440**. Each of the components **410**, **420**, **430**, and **440** are interconnected using a system bus **450**. The processor **410** is capable of processing instructions for execution within the controller **400**. The processor may be designed using any of a number of architectures. For example, the processor **410** may be a CISC (Complex Instruction Set Computers) processor, a RISC (Reduced Instruction Set Computer) processor, or a MISC (Minimal Instruction Set Computer) processor.
- (53) In one implementation, the processor **410** is a single-threaded processor. In another implementation, the processor **410** is a multi-threaded processor. The processor **410** is capable of processing instructions stored in the memory **420** or on the storage device **430** to display graphical information for a user interface on the input/output device **440**.
- (54) The memory **420** stores information within the controller **400**. In one implementation, the memory **420** is a computer-readable medium. In one implementation, the memory **420** is a volatile memory unit. In another implementation, the memory **420** is a non-volatile memory unit.
- (55) The storage device **430** is capable of providing mass storage for the controller **400**. In one implementation, the storage device **430** is a computer-readable medium. In various different implementations, the storage device **430** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device.
- (56) The input/output device **440** provides input/output operations for the controller **400**. In one implementation, the input/output device **440** includes a keyboard and/or pointing device. In another implementation, the input/output device **440** includes a display unit for displaying graphical user interfaces.
- (57) The features described can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. The apparatus can be implemented in a computer program product tangibly embodied in an information carrier, for example, in a machine-readable storage device for execution by a programmable processor; and method steps can be performed by a programmable processor executing a program of instructions to perform functions of the described implementations by operating on input data and generating output. The described features can be implemented advantageously in one or more computer programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device. A computer program is a set of instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.
- (58) Suitable processors for the execution of a program of instructions include, by way of example, both general and special purpose microprocessors, and the sole processor or one of multiple

processors of any kind of computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memories for storing instructions and data. Generally, a computer will also include, or be operatively coupled to communicate with, one or more mass storage devices for storing data files; such devices include magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and optical disks. Storage devices suitable for tangibly embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, such as EPROM, EEPROM, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, ASICs (application-specific integrated circuits).

(59) To provide for interaction with a user, the features can be implemented on a computer having a display device such as a CRT (cathode ray tube) or LCD (liquid crystal display) monitor for displaying information to the user and a keyboard and a pointing device such as a mouse or a trackball by which the user can provide input to the computer. Additionally, such activities can be implemented via touchscreen flat-panel displays and other appropriate mechanisms.

(60) The features can be implemented in a control system that includes a back-end component, such as a data server, or that includes a middleware component, such as an application server or an Internet server, or that includes a front-end component, such as a client computer having a graphical user interface or an Internet browser, or any combination of them. The components of the system can be connected by any form or medium of digital data communication such as a communication network. Examples of communication networks include a local area network ("LAN"), a wide area network ("WAN"), peer-to-peer networks (having ad-hoc or static members), grid computing infrastructures, and the Internet.

(61) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of features specific to particular implementations. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub combination or variation of a subcombination.

(62) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(63) Advantageously, the systems and processes described herein may allow for realistic emulation of the effect of brine composition on the interfacial tension measurement, and realistic emulation of the effect of additives on interfacial phenomena. The systems and methods may eliminate uncertainties due to multiple separated runs and contamination between runs, may be efficient with respect to experimental time management, and may minimize the amount of solvents and chemicals used for cleaning that follows each single run.

(64) While the disclosure includes a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments may be devised which do

not depart from the scope of the present disclosure. Accordingly, the scope should be limited only by the attached claims.

Claims

1. A method for determining interfacial tension of a hydrocarbon in a brine fluid, comprising: injecting a first brine fluid into a test cell, the first brine fluid having an initial ionic composition; injecting a hydrocarbon fluid into the test cell; contacting the hydrocarbon fluid with the first brine fluid, forming a droplet; measuring the interfacial tension of the hydrocarbon fluid in contact with the first brine fluid; at least partially displacing the first brine fluid with an inert gas; measuring a ionic composition of the displaced first brine fluid in an ionic chromatograph; and comparing the measured ionic composition to the initial ionic composition.
 2. The method of claim 1, wherein when the measured ionic composition matches the initial ionic composition, the method further comprising: injecting a second brine fluid into the test cell, the second brine fluid having a second initial ionic composition; injecting the hydrocarbon fluid into the test cell; contacting the hydrocarbon fluid with the second brine fluid, forming a droplet; measuring the interfacial tension of the hydrocarbon fluid in contact with the second brine fluid; at least partially displacing the second brine fluid with the inert gas; measuring a ionic composition of the displaced second brine fluid in an ionic chromatograph; and comparing the measured ionic composition to the second initial ionic composition.
 3. The method of claim 1, wherein when the measured ionic composition does not match the initial ionic composition, the method further comprising: circulating the first brine fluid into the test cell; at least partially displacing the first brine fluid with the inert gas; measuring the ionic composition of the displaced first brine fluid in the ionic chromatograph; and comparing the measured ionic composition to the initial ionic composition.
 4. The method of claim 1, further comprising filtering the at least partially displaced first brine fluid in a filter prior to measuring the ionic composition.
 5. The method of claim 2, wherein when the second measured ionic composition matches the second initial ionic composition, the method further comprising: injecting a third brine fluid into the test cell, the third brine fluid having a third initial ionic composition; injecting the hydrocarbon fluid into the test cell; contacting the hydrocarbon fluid with the third brine fluid, forming a droplet; measuring the interfacial tension of the hydrocarbon fluid in contact with the third brine fluid; at least partially displacing the third brine fluid with the inert gas; measuring a ionic composition of the displaced third brine fluid in the ionic chromatograph; and comparing the measured ionic composition to the third initial ionic composition.
 6. The method of claim 2, wherein when the measured ionic composition does not match the second initial ionic composition, the method further comprising: circulating the second brine fluid into the test cell; at least partially displacing the second brine fluid with the inert gas; measuring the ionic composition of the displaced second brine fluid in the ionic chromatograph; and comparing the measured ionic composition to the second initial ionic composition.
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